Email Linkedin

Murdock Aubry

GitHub Website

Education

University of Toronto **December 2026 (Expected)**

MSc in Applied Computing, Applied Mathematics stream

University of Toronto **June 2024**

HBSc Mathematics and Physics Specialist. High distinction.

Coursework: Machine learning, Deep learning, numerical analysis, optimization, real analysis, complex analysis, PDEs, linear algebra, probability, topology, differential geometry, theoretical physics.

Experience

Faculty Affiliate Researcher · Vector Institute **November 2023 - Present**

- Supervised by Professor Vardan Papyan.
- Research deep learning architectures with a focus on interpretability of LLMs.
- Discovered phenomenon called Transformer Alignment (TA), and displayed correlation between TA and generalization.
- Submitted work to ICML and NeurIPS 2024. See a preprint here.

Research Assistant · University of Toronto **April 2023 - Present**

- Developing algorithms for solving scalar differential equations and for numerical integration.
- Our methods dominate over existing methods, particularly in the high-frequency regime.
- Establish theoretical framework, implement, test and iteratively optimize, publish results in academic journals.
- This work has lead to a preprint.

Research Assistant · CERN **September 2023 - April 2024**

- Supervised by Professor Nikolina Ilic, collaborated with her CERN research team of graduating students.
- Developing a machine learning algorithm to detect tau neutrinos and their decay products.
- I combatted an imbalanced dataset by implementing a modified, weighted loss function.
- My contributions vastly improved the performance of the model, reaching historically minimal classification error.
- See here for up to date progress.

Teaching Assistant · University of Toronto **September 2023 - Present**

- Fall 2024: MAT292, ODEs for second year engineering students.
- Winter 2024: MAT187, Calculus 2 for first year engineering students.
- Fall 2023: MAT244, ODEs for second year math major students.

Tutor · Forest Hill Tutoring **April 2023 - Present**

- High school math and physics tutoring, largely focused on helping grade 11 and 12 IB students.
- Principal instructor for Ontario credited courses such as MDM4U, MHF4U and MCV4U.

Civil Engineering Assistant · SKYGRiD **April 2022 - October 2022**

- Worked on a 57 storey building in downtown Toronto assisting the senior and assistant supervisors
- Independently managed construction and renovation of small townhouse in downtown Toronto.
- Worked alongside mechanical, civil, and electrical engineers throughout entire construction process

Awards and Scholarships

- **VSIA.** Vector Scholarship in Artificial Intelligence. Valued at $17,500. **2024**

Academia

- The Levin approach to the numerical calculation of phase functions. **Murdock Aubry**, James Bremer. Preprint.
- Transformer Alignment in Large Language Models. **Murdock Aubry**, Hoaming Meng, Anton Sugolov, Vardan Papyan. Preprint.
- Worked alongside mechanical, civil, and electrical engineers throughout entire construction process
- Grad Applied^2 Math Seminar 2023. The emergence of clusters in self-attention dynamics.
- DUNE Talk 2023. Development of graph neural networks for tau neutrino detection.
- CUMC Talk 2023. The Adaptive Levin Method.

Technical Skills

- **Languages.** Python, R, Fortran, HTML, CSS, SQL, Bootstrap, Bash, Mathematica, MATLAB, LaTeX.
- **Packages.** PyTorch, Transformers, TensorFlow, Numpy, Scipy, Pandas, Matplotlib, Seaborn, JSON, OS, Selenium, BeautifulSoup.
- **Software.** AutoCAD, macOS, Git, SSH, SLURM, Tensorboard .